

14. Determine whether $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is onto if

a) $f(m, n) = 2m - n$

Solution: Let, $f(m, n) = x$ where $x \in \mathbb{Z}$.
 $\therefore (2m - n) = x$. If $m = n = x$, then $2m - n = 2x - x = x$

$\therefore \forall x \in \mathbb{Z}, \exists m, n \in \mathbb{Z}$ s.t. $f(m, n) = x$. By Defn. 7, f is onto $\square$

b) $f(m, n) = m^2 - n^2$

Solution: Let, $f(m, n) = 2$ where $2 \in \mathbb{Z}$.

$$\therefore m^2 - n^2 = 2 \Rightarrow m^2 = n^2 + 2 \Rightarrow m = \sqrt{n^2 + 2}$$

By Defn. 1 from Sec-1.7, an integer can either be even or odd.
The following 4 cases arise:

Case 1: m is even, n is even. By Defn. 1 from Sec-1.7, $\exists$ integers n_1 and n_2 s.t.
 $m = 2n_1$ and $n = 2n_2$. $m^2 - n^2 = (2n_1)^2 - (2n_2)^2 = 4(n_1^2 - n_2^2)$

$\therefore (m^2 - n^2)$ is always a multiple of 4 here. 2 is not a multiple of 4.

Case 2: m is even, n is odd. By Defn. 1 from Sec-1.7, $\exists$ integers n_1 and n_2 s.t.
 $m = 2n_1$ and $n = 2n_2 + 1$. $m^2 - n^2 = (2n_1)^2 - (2n_2 + 1)^2 = 4(n_1^2 - n_2^2 - n_2) - 1$
 $= 2(2n_1^2 - 2n_2^2 - 2n_2 - 1) - 1$

By Defn. 1 from Sec-1.7, $m^2 - n^2$ is odd $\because 2n_1^2 - 2n_2^2 - 2n_2 - 1 \in \mathbb{Z}$.

Case 3: m is odd, n is even. Similar to Case 2

Case 4: m is odd, n is odd. By Defn. 1 from Sec-1.7, $\exists$ integers n_1 and n_2 s.t.
 $m = 2n_1 + 1$ and $n = 2n_2 + 1$. $m^2 - n^2 = (2n_1 + 1)^2 - (2n_2 + 1)^2 = (4n_1^2 + 4n_1 + 1) - (4n_2^2 + 4n_2 + 1)$
 $= 4(n_1^2 + n_1 - n_2^2 - n_2)$

$(m^2 - n^2)$ is a multiple of 4. 2 is not a multiple of 4.

$\therefore \forall m, n \in \mathbb{Z} f(m, n) \neq 2$. By Defn. 7, f is not onto $\square$